

Beyond accuracy: a review of the robustness and operational relevance of ML/DL-based ADS-B spoofing detection

Jusuf Dacić¹, Nejra Kapidžija¹

¹Faculty of Traffic and Communications, University of Sarajevo, Zmaja od Bosne 8, Sarajevo 71000, Bosnia and Herzegovina

Abstract

The absence of built-in authentication in the Automatic Dependent Surveillance-Broadcast (ADS-B) protocol enables false-message injection and can undermine surveillance integrity. Machine learning and deep learning methods can detect such manipulations, but high experimental performance does not by itself establish operational reliability. This paper reviews 19 publications from 2021-2026 identified through searches of IEEE Xplore, ScienceDirect, Google Scholar, and reference tracking. Five research questions guide the analysis of data and attack provenance, adversarial testing, false alarms, timing metrics, generalization, and evaluation gaps. Among 18 papers with established data provenance, 17 use real flights with constructed anomalies or attacks, while one is simulation-based; one additional paper has unclear provenance. No confirmed operational spoofing attack was identified in the analyzed data. None of the 17 detector-proposing papers reports adversarial testing of the proposed detector. Ten report false-alarm indicators, six report computational timing, and one reports messages to detection. External spatial evaluation is identified in one paper, with two additional partial or indirect approaches to generalization. The review identifies priorities for transparent testing protocols, representative benign data, realistic threat models, false-alert control, and external validation. Reported accuracy alone does not establish the operational relevance of these detectors.

Keywords: ADS-B, spoofing, machine learning, deep learning, adversarial robustness, generalization

1 Introduction

Primary surveillance radar determines the range and azimuth of a target based on the reflected radio signal. Secondary surveillance radar (SSR) uses responses from the aircraft transponder: the ground system determines range and azimuth, while the transponder transmits an identification code and, depending on the mode, barometric altitude data. In ADS-B (Automatic Dependent Surveillance-Broadcast), the aircraft automatically broadcasts its own position and movement data. Position is usually obtained from a global navigation satellite system (GNSS), while altitude data may include barometric and geometric altitude [1].

ADS-B Out denotes the transmission of data, and ADS-B In their reception and use. The

availability of surveillance depends on the receiving infrastructure. Coverage of oceanic and remote areas may involve satellite receivers [2]. Equipage requirements are not universal. In the US, the mandate has applied since January 2020 for designated airspace, while the European framework distinguishes categories of aircraft and transitional deadlines [3], [4].

ADS-B messages are, by default, broadcast without cryptographic sender authentication. An attacker can therefore attempt to inject false messages, impersonate an aircraft's identity, or alter the displayed trajectory. The success of an attack depends on access to the communication channel or data stream, the receiver, and additional verification mechanisms, so not every false message becomes an accepted target in the operational system [5]-[7].

Corresponding author: Jusuf Dacić (jusuf.dacic@fsk.unsa.ba)

© 2026 The Author(s). This work is licensed under a Creative Commons Attribution 4.0 International License.

In this paper, spoofing encompasses false representation and manipulation of ADS-B information, in particular the injection of false targets and the alteration of movement data. Jamming of the radio channel is not, by itself, spoofing. A distinction must also be made between falsifying an ADS-B message and deceiving the GNSS receiver that supplies navigation data to the aircraft. An anomaly detector recognizes a deviation from the expected pattern, but the deviation alone does not establish the cause: possible causes include an attack, a sensor error, message loss, or an unusual legitimate maneuver [7], [8].

Cryptographic authentication and anomaly detection play complementary roles. The former verifies the origin and integrity of messages, while the latter can detect unusual behavior, including some manipulations that pass preventive controls. The introduction of protection must take into account protocol compatibility, key management, and the requirements of existing avionics [5], [7].

The aim of the review is to assess what evidence existing machine learning (ML) and deep learning (DL) methods provide regarding the reliability of ADS-B spoofing detection. Here, ML includes classical statistical and ensemble methods, and DL includes multi-layer neural architectures. The object of assessment is the quality and scope of reported evaluation, not the certification of detectors or the ranking of architectures according to a single metric. Operational relevance denotes the extent to which a reported result can be expected to hold in real surveillance: with representative traffic, a realistic attacker, a tolerable alert load, and the required

detection delay. The overarching question is what evidence existing ML/DL detectors provide about their robustness and operational relevance beyond reported accuracy. To answer it, five research questions (RQs) were identified:

- RQ1: What is the provenance of the benign data and attacks used in the evaluation?
- RQ2: To what extent do the papers examine deliberate manipulations of the model and defenses against them?
- RQ3: Which false-alarm and timing-performance indicators are reported?
- RQ4: What is the evidence of generalization to routes, regions, and attacks outside the training conditions?
- RQ5: Which evaluation gaps limit the assessment of operational applicability?

Table 1 summarizes the related literature and positions this review. The reviews [5]-[7] describe ADS-B threats, protection methods, and open problems, and already discuss adversarial threats, false alerts, and application limitations. This paper instead follows the five RQs through individual experiments and distinguishes papers that propose a detector, papers that test attacks, and defense experiments. These roles may overlap: Luo et al. [9] propose an attack, but also examine adversarial training of existing models. STLC-Net [8], which includes external evaluation under different regional and reception conditions but uses rule-derived anomaly labels rather than constructed or confirmed spoofing attacks, serves only as context for generalization and label validity, outside the statistics of the 19-paper corpus.

Table 1. Related literature and positioning of this review

Work	Focus	Relation to this review
Çevik and Akleylek [6]	Systematization of ML/DL anomaly detection for ADS-B	Method-level overview, not per-experiment criteria
Ahmed et al. [5]	ADS-B threats, existing solutions, future research	Broad security scope; detection is one solution class
Shi et al. [7]	ADS-B and Remote ID attacks, detection, countermeasures	Wider system security; ML/DL detection is one part
Luo et al. [9]	Gradient-based attack on detectors; adversarial training	In the corpus as an attack-focused paper
Zhong et al. [8]	Kinematic anomaly detection with rule-derived labels	Context for generalization; outside the corpus
This paper	Evaluation evidence across five RQs	Criterion-based analysis of 19 papers

The contribution of this paper is a joint criterion-based analysis of the selected corpus and the linking of the observed limitations to concrete recommendations for evaluation. Counts of papers describe reporting practice within this corpus, not the reliability of individual detectors.

2 Review methodology

2.1 Sources and search strategy

The search covered IEEE Xplore, ScienceDirect, and Google Scholar, with reference tracking of relevant publications. The search was focused on the period 2021-2026. The final check of the core corpus was carried out in September 2026. Older publications were used to explain the development of the field, without being included in the quantitative analysis.

The main query was:

(“ADS-B”) AND (“spoofing” OR “anomaly detection” OR “intrusion detection”) AND (“machine learning” OR “deep learning”)

Supplementary searches covered the following terms:

- “adversarial ADS-B” and “adversarial attack” AND “ADS-B”
- “ADS-B” AND “false alarm rate” and “ADS-B” AND “real-time detection”
- “ADS-B” AND “generalization”.

Table 2 shows the recorded coverage of the main search by source. Estimates of the total number of Google Scholar hits were not used as the number of unique papers reviewed.

Table 2. Recorded coverage of the main search (September 2026)

Source	Results returned	Reviewed range
IEEE Xplore	41	All titles
ScienceDirect	134	All titles
Google Scholar	Approximately 2,200	First 80 results by relevance

The stated main query was used for all three sources, with a publication period of 2021-2026.

2.2 Inclusion criteria and selection

Included were papers in English, published in a peer-reviewed journal or conference proceedings from 2021 to 2026, with an available full text and an experimental evaluation of ML/DL detection of ADS-B manipulations or of attacks on such detectors. An available author version of an accepted conference paper could be used for reading, with the publication status noted. Excluded were reviews without their own evaluation, theses, conceptual proposals without an experiment, papers outside the period, solutions relying exclusively on cryptographic, hardware, or jamming measures without a corresponding ML/DL evaluation, and studies of kinematic or operational anomalies without manipulation or attack labels.

The main-search record contains 41 results from IEEE Xplore, 134 from ScienceDirect, and 80 reviewed Google Scholar results. In total, 255 records were reviewed. After removing 13 duplicates, 242 unique titles remained. Initial screening of titles by topical scope excluded 167 records, leaving 75 papers for more detailed screening of titles and abstracts. Six papers found through reference tracking were added, forming a set of 81 papers. Relevant titles from the supplementary searches were merged into the same screening. Separate counts per supplementary query are not shown.

Of the 81 papers, 32 were excluded after screening of titles and abstracts. For 15 of the remaining 49, the full text could not be obtained, and five were excluded due to their year of publication before the full-text assessment. Twenty-nine full texts were assessed. Ten were excluded: three reviews, one conceptual framework without an experiment, two papers outside the topical scope, one paper in another language, and three papers published before 2021. Because the Google Scholar year filter is not fully reliable, some earlier papers appeared among its results; the publication year of each paper was therefore verified manually, which led to the eight period-based exclusions. The final corpus comprises 19 papers. Of the six papers found through reference tracking, five were included. The foundational papers by Habler and Shabtai [10] and Ying et al. [11] belong to older, contextual literature.

Fig. 1 shows the stages of searching, duplicate removal, initial title screening, and detailed selection. The diagram was prepared by the

authors from the search record, following the transparent-reporting principle of Preferred

Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 [12].

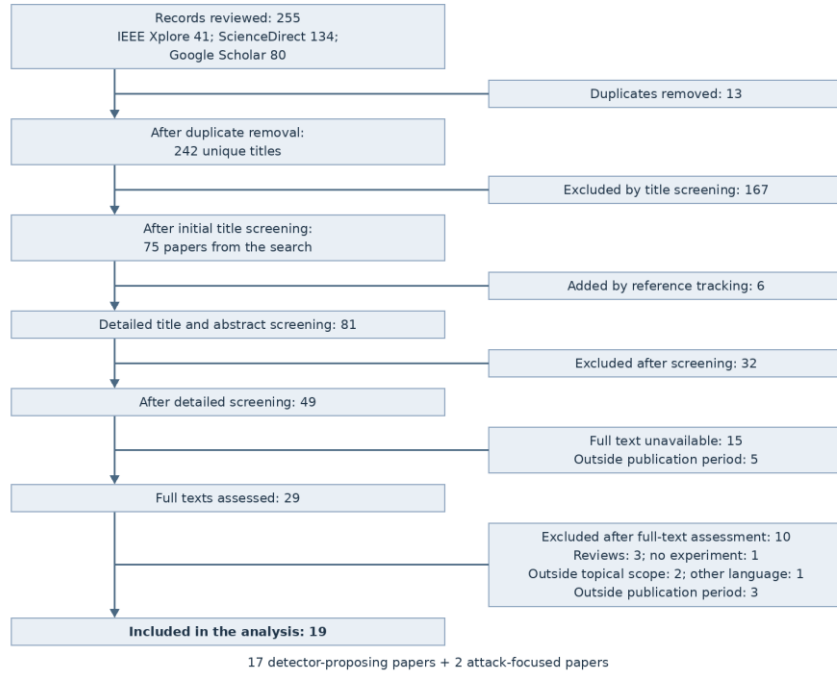


Fig. 1. Flow of the search and selection of papers

2.3 Extraction and assessment rules

For each included paper, the method, the provenance of the evaluation data, the type of attack, the reported metric, and the evidence for the five criteria from Table 3 were recorded. The unit of analysis is the publication, not the individual model, so a paper comparing several models is counted once. Titles and abstracts were screened independently by both authors, and the final inclusion decisions were agreed jointly. Both

authors extracted the data independently, applying the recording rules in Table 3 as a shared coding scheme, and the final classification was established by consensus. Because of differing tasks, datasets, and protocols, the results are synthesized descriptively, without a common average accuracy or a meta-analysis. No composite quality score was assigned: the five criteria record the evaluation properties relevant to the RQs, and a single score would hide which of them is missing.

Table 3. Criteria and recording rules

Criterion	Evidence recorded
Data and attack provenance	Real or simulated benign traffic, method of anomaly construction, and separate existence of a confirmed operational attack.
Adversarial evaluation	Deliberate attack on the model: evasion, poisoning, or another appropriate scenario. Defense is recorded separately.
False alarms	False-positive rate (FPR), false-alarm rate (FAR), or an equivalent indicator, e.g. specificity. The positive class and decision unit matter.
Timing performance	Separately: computational processing/detection time and the number of messages from the start of the attack to detection. Inference latency and training time are not the number of messages to detection.
Generalization	External test on other routes or regions; narrower or indirect checks are recorded separately, with the type of evidence stated. Architecture alone is not an external test.

A criterion is counted as met only if the experimental description reports it; an unreported test does not show that the method would fail it. Reporting of specificity is counted as an indicator of the FPR because, with the same positive class, $FPR = 1 - \text{specificity}$. Architectural intent, without an external test, is not counted as empirical generalization, and a test on an unseen attack type is not counted as adversarial evaluation unless it deliberately targets the learned model.

The criterion-based assessment of detectors uses a denominator of 17. The two papers focused on attacks [9], [13] are analyzed separately. Data provenance is considered for all 19 papers, with the unclarified case labeled separately.

2.4 Review limitations

Scopus and Web of Science were not searched, and Google Scholar was reviewed down to the first 80 ranked results, so relevant papers may have been omitted. Dates are recorded at the month level, and the complete queries adapted to

each platform and the separate results of the supplementary searches are not shown in full. For this reason the search is not fully reproducible at the level of every returned record.

Full texts were sought through institutional access, but 15 could not be obtained. Their unavailability may affect the composition of the corpus and the aggregate results; since at least one of them is attack-focused, the adversarial findings apply to the analyzed corpus. Individual papers also differ in the level of detail of reporting and may use related datasets, so the number of publications is not the number of independent confirmations.

3 Overview of analyzed approaches

The corpus comprises 19 publications: 17 proposing detectors [14]-[30] and two focused on attacks on models [9], [13]. Table 4 groups them by methodological family. The grouping serves the organization of the presentation; the families are not mutually exclusive, since sequential models can simultaneously be autoencoders.

Table 4. Methodological families in the analyzed corpus

Group	Papers	Representative methods
Sequential and one-class anomaly detection models	[16], [18], [20], [24]	Stacked long short-term memory (LSTM) encoder-decoder [20]; transformer encoder and extended LSTM (xLSTM) with transfer learning [24]; differencing with a non-recurrent autoencoder [18]; Isolation Forest and one-class support vector machine (SVM) on physical-layer features [16]
Classical and ensemble models	[15], [17], [22], [25], [27], [28]	k-nearest neighbors (kNN), SVM, decision trees, random forests, boosting; optimized CatBoost [27]; meta-classifier [28]; lightweight multi-attack models [15]
Hybrid and specialized approaches	[14], [19], [21], [23], [26], [29], [30]	Tabular deep learning [14]; spatial, temporal, and radio-frequency features [19]; false-data generator [21]; hybrid quantum network [23]; per-attack algorithms [26]; detection with data recovery [29]; time-frequency processing [30]
Attacks on models and defense	[9], [13]	Poisoning data from an improved generative adversarial network (GAN) [13]; Time Neighborhood Accumulation Iteration Fast Gradient Sign Method (TNAI-FGSM), attack transferability, and adversarial training [9]

Reconstruction-based approaches learn expected patterns, while the physical layer provides another type of information about message authenticity [16]. These approaches rest on different assumptions about sensor infrastructure and the attacker’s position. Testing in [24] also covers a previously unseen attack type. Good classification performance of classical and ensemble methods does not remove the need

for separate checks of data splitting, false alerts, and transfer to another context.

In the attack-focused papers, evasion manipulates the input at the time of the model’s use, while poisoning affects the data or its learning process. The use of a GAN in [13] is not in itself the reason for its classification; the paper counts as an attack because the generated data are used to poison model training.

4 Robustness and limitations analysis

Table 5 classifies all 19 papers by the criteria from Table 3, and its last row gives the totals, over all 19 papers for data provenance and over the 17 detector-proposing papers for the other criteria. Data denotes data provenance: R marks real flights with constructed attacks, S a simulation-based or synthetic dataset, and U unclear provenance. Adv., FA, Tim., and Gen. denote adversarial evaluation, false alarms, timing performance, and generalization. E marks a met criterion, P partial or indirect evidence, NI no reported evidence, N/A a criterion that does not apply, A an attack on the model, D a defense, and E* the number of messages from the start of the attack to detection.

Table 5. Criterion-based classification of the analyzed papers

Paper	Data	Adv.	FA	Tim.	Gen.
[14]	R	NI	NI	NI	NI
[15]	R	NI	NI	E	NI
[16]	R	NI	NI	NI	NI
[17]	R	NI	E	NI	NI
[18]	R	NI	E	E*	E
[19]	U	NI	E	NI	NI
[20]	R	NI	NI	NI	P
[21]	R	NI	E	NI	NI
[22]	R	NI	E	NI	NI
[23]	R	NI	NI	NI	NI
[24]	R	NI	E	E	P
[25]	R	NI	E	E	NI
[26]	R	NI	NI	NI	NI
[27]	R	NI	E	E	NI
[28]	R	NI	NI	NI	NI
[29]	R	NI	E	E	NI
[30]	R	NI	E	E	NI
[9]	R	A, D	N/A	N/A	N/A
[13]	S	A	N/A	N/A	N/A
Total	17 R, 1 S, 1 U	0	10	7	1 E, 2 P

Of the 18 papers with clarified provenance, 17 use real flights with constructed anomalies or

attacks and one is simulation-based; for one paper the provenance is not sufficiently clarified. The share of papers using real flights is therefore 17/18, i.e. 94.4%. This share does not measure attack realism, because no confirmed spoofing event from operational traffic was identified in the analyzed datasets. In [9], for example, OpenSky data from 50 real flights was used for training and 40 flights with simulated anomalies for testing. Habler and Shabtai [20] also examine a real route deviation, an operational anomaly without a confirmed malicious cause. The lack of public, confirmed, and labeled attack examples remains the main obstacle to validating detectors on realistic threats.

None of the 17 detector-proposing papers tests deliberate manipulations targeting the model itself. Adversarial research enters the corpus through the attack-focused papers [9], [13], and paper [9] also tests adversarial training as a defense. The earlier paper by Shang et al. [31] provides historical context but is outside the 2021-2026 statistics.

Table 8 in [9] reports for LSTM an accuracy of 90.09% before the attack, 51.44% after the TNAI-FGSM attack, and 75.43% after adversarial training. Fig. 2, the authors’ own chart based on these values from [9], shows a partial recovery without a return to the initial value. Because normal and anomalous data each form half of the test data in [9], 51.44% is close to chance level, and the perturbations mainly cause normal data to be flagged as anomalous. An adversarial attack can therefore also act as a source of false alarms, which links RQ2 and RQ3. The defense helps, but the experiment does not cover adaptive attacks that take the defense into account.

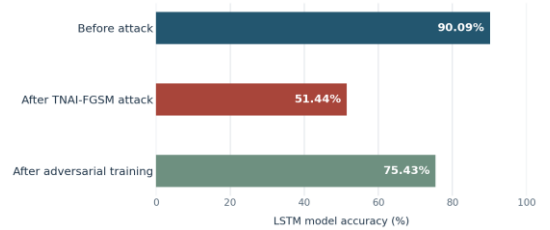


Fig. 2. Attack and partial recovery of the LSTM detector

In ten of the 17 detector papers, FPR, FAR, or an equivalent indicator is recorded (Table 5). Reporting differs by positive class, threshold, and decision unit, so FAR and FPR values are not directly interchangeable. Rahman et al. [27], for example, report specificity, while Yue et al. [29]

state FPR and the false-negative rate (FNR). Ngamboé et al. [24] present FPR in the results tables, while Ould Slimane et al. [25] state the false-alarm probability in Table IV. Karam et al. [21] report FPR in Table III, but its textual description conflates the denominators of FPR and the proportion of false alerts. Paper [21] is still counted, so the ten papers do not form ten directly comparable measurements.

$$FPR = FP / (FP + TN) \quad (1)$$

where TP, FP, TN, and FN denote the numbers of true positives, false positives, true negatives, and false negatives.

If the attack is the positive class, the false-positive rate is given by Eq. (1), and precision is $TP / (TP + FP)$. With an unchanged model, threshold, and conditional data distributions, a change in the share of attacks need not change FPR. However, a decrease in the share of attacks can strongly reduce precision, that is, the proportion of correct alerts among all alerts [32].

Fig. 3 shows the authors' own illustrative calculation. Out of one million messages, 1,000 are attacks. With a recall of 99% and an FPR of 0.1%, 990 true positive and 999 false positive decisions are expected, and precision is approximately 49.8%. The values are not an estimate of real attack frequency or a result of any analyzed detector. The calculation assumes a separate decision for each message; an operational system can aggregate messages into alerts, so controller workload requires measurement at the level of events and operating time.

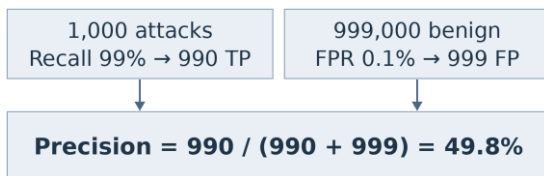


Fig. 3. Illustrative alert precision at a 0.1% attack share

To assess usability, FPR should be reported on representative benign data, precision and recall at clearly stated attack shares, and the number of false alerts per hour or per flight. A zero observed rate on a finite test set does not prove that the true rate is zero, so sample size and uncertainty estimates are needed. The decision threshold must be chosen without using the test labels.

Six papers report computational timing [15], [24], [25], [27], [29], [30], and Fried and Last [18] measure the number of messages from the start of the attack to detection, so seven papers report a timing indicator.

Ngamboé et al. [24] report, for known attack types, an inference latency of 7.26 s for xLSTM and 2.1 s for the transformer (Table V of the source). The authors describe it as the interval from receipt of a message to its verification and compare it with the SSR refresh interval of 5-12 s, which provides a concrete operational reference. It does not represent the measured number of messages from the start of the attack to detection, so this paper is counted only under computational timing. Ould Slimane et al. [25] report a C-SVM detection time of 0.6242 s, but not the number of messages to detection, so it is also counted under computational timing. The processing time of a single window, the processing time of a batch of messages, and the duration from the start of an attack to an alert represent different quantities.

External spatial evaluation is recorded in one paper, which examines routes and a region outside the training set [18]. Two papers address generalization partially or indirectly: a check of an operational deviation [20] and a test on an unseen attack [24]. The architectural mechanism for distribution shift in [30] is not counted, because it is not accompanied by an external test.

Fried and Last [18] achieve a receiver operating characteristic area under the curve (ROC AUC) of 0.935-0.951 on flights outside the training set, versus 0.627 for the compared existing methods in the same evaluation context. This supports the usefulness of the proposed representation for new routes within the tested scenarios.

Random message-level splitting can place closely related samples from the same flight or overlapping windows into both the training and the test set, which is a possible source of data leakage to be checked in each paper. It is preferable to split by flights, time, and regions, and to fit normalization, feature selection, and thresholds only on the training and validation sets.

Twelve of the 17 detector papers report a headline accuracy or F1-score above 95% [14], [15], [17], [19], [21], [22], [24], [25], [27]-[30], while [23] reports the lowest values, 90.17-94.05%. Such values should not be converted into

a single ranking, because direct comparison requires common data, positive class, decision unit, threat model, threshold-selection rule, and resource constraints. Without these, a high reported value primarily describes success in a particular experiment. The gaps summarized in Table 5 define the recommendations in Section 5.

5 Recommendations for future studies

Table 6 links the gaps identified in the robustness and limitations analysis to recommendations for future studies. The adversarial-training experiment in [9] represents a starting point for comparison, but not a universal confirmation of security. The acceptability of detection delay is assessed against the requirements of the specific surveillance function, including the role of the operator and other surveillance sources.

Table 6. Recommendations for future studies

Area	Recommended practice
Data and threat model	Separate the provenance of benign data from the attack construction; describe routes, regions, receivers, time span, and labels. State the attacker’s access, knowledge, and physical constraints. Distinguish constructed attacks, real anomalies of unknown cause, and confirmed incidents; use diverse, plausibility-checked scenarios.
Adversarial robustness	Use attacks tailored to the threat model, including different attacker knowledge and poisoning. Report attack success with the change in recall, FPR, and benign performance; test each defense against an attack that accounts for it.
Alerts and timing	Measure false alerts per hour, delay to a confirmed event, and alert aggregation; report median and high-percentile delay, throughput under load, and hardware.
Generalization and reproducibility	Describe the data split; test other routes, periods, reception conditions, or regions with fixed parameters; report unseen attacks separately; share code, data versions, and variability across repeated runs.

6 Conclusion

A review of 19 publications from the period 2021-2026 shows the diversity of ML/DL approaches to detecting ADS-B manipulations and an uneven scope of their evaluation. The answers to the research questions are as follows.

RQ1: Seventeen of 18 papers with clarified data provenance combine real flights with constructed anomalies or attacks. One is recorded as simulation-based. For one paper the provenance remains unclarified. No confirmed operational spoofing attack was identified in the analyzed data, so real benign traffic is not evidence of attack realism.

RQ2: Among the 17 detector-proposing papers, no separate adversarial testing was recorded under the applied criterion. Two other papers examine attacks on models, and one of them also examines adversarial training as a defense.

RQ3: In ten of the 17 detector papers, a false-alarm indicator is recorded, with differences and ambiguities in the definitions. Six report a computational timing metric, and a seventh paper reports the number of messages from the start of the attack to detection. These indicators are not interchangeable and do not show that operational timing requirements are met.

RQ4: External spatial evaluation is recorded in one paper. Two additional papers consider a real anomaly or a previously unseen attack. The architectural mechanism for distribution shift in [30] is not counted, because it lacks an external test.

RQ5: The assessment of operational applicability is limited by the representativeness of attack scenarios, uneven metrics, sparse external evaluations, and insufficiently comparable timing protocols. Priorities are clear threat models, leakage-free data splits, evaluation of adaptive attacks, false alerts per hour or per flight, and delay measurement under representative load.

The reported results support the potential of the methods under the tested conditions, but do not allow a universal ranking or confirmation of operational readiness. The conclusions are limited to the analyzed corpus.

References

- [1] Federal Aviation Administration (FAA), "Existing surveillance technologies for Upper Class E Traffic Management (ETM), version 2.2," Jul. 26, 2023. [Online]. Available: https://www.faa.gov/uas/advanced_operations/upper_class_etm/ETM-Surveillance-Technologies.pdf. [Accessed: Sep. 14, 2026].
- [2] EUROCONTROL, "Automatic dependent surveillance broadcast (ADS-B)," [Online]. Available: <https://www.eurocontrol.int/service/automatic-dependent-surveillance-broadcast>. [Accessed: Sep. 14, 2026].
- [3] Federal Aviation Administration (FAA), "FAA issues policy for ADS-B compliance," Apr. 2, 2019. [Online]. Available: <https://www.faa.gov/newsroom/faa-issues-policy-ads-b-compliance>. [Accessed: Sep. 14, 2026].
- [4] European Union Aviation Safety Agency (EASA), "Amendment to the airspace requirements on ADS-B and Mode S - important information for third country operators (TCO)," 2020. [Online]. Available: <https://www.easa.europa.eu/en/newsroom-and-events/news/amendment-airspace-requirements-ads-b-and-mode-s>. [Accessed: Sep. 14, 2026].
- [5] W. Ahmed, A. Masood and J. Manzoor, "Towards secure air traffic surveillance: A survey on ADS-B threats, existing solutions, and future research," *IEEE Open Journal of the Computer Society*, vol. 7, pp. 290-313, Jan. 2026. <https://doi.org/10.1109/OJCS.2026.3651384>
- [6] N. Çevik and S. Akleylek, "SoK of machine learning and deep learning based anomaly detection methods for automatic dependent surveillance-broadcast," *IEEE Access*, vol. 12, pp. 35643-35662, Feb. 2024. <https://doi.org/10.1109/ACCESS.2024.3369181>
- [7] Q. Shi, T. D. Caleb, S. Shao and N. Kaabouch, "Security of ADS-B and Remote ID systems: Cyberattacks, detection techniques, and countermeasures," *Sensors*, vol. 26, no. 2, p. 634, Jan. 2026. <https://doi.org/10.3390/s26020634>
- [8] L. Zhong, X. Liu, K. Xu and W. Pan, "STLC-Net: A short-window temporal latent-consistency network for ADS-B trajectory anomaly detection," *Aerospace*, vol. 13, no. 8, p. 706, Aug. 2026. <https://doi.org/10.3390/aerospace13080706>
- [9] P. Luo, B. Wang, J. Tian, C. Liu and Y. Yang, "Adversarial attacks against deep-learning-based ADS-B unsupervised anomaly detection models in the context of air traffic management," *Sensors*, vol. 24, no. 11, p. 3584, Jun. 2024. <https://doi.org/10.3390/s24113584>
- [10] E. Habler and A. Shabtai, "Using LSTM encoder-decoder algorithm for detecting anomalous ADS-B messages," *Computers & Security*, vol. 78, pp. 155-173, Sep. 2018. <https://doi.org/10.1016/j.cose.2018.07.004>
- [11] X. Ying, J. Mazer, G. Bernieri, M. Conti, L. Bushnell and R. Poovendran, "Detecting ADS-B spoofing attacks using deep neural networks," in *IEEE Conference on Communications and Network Security (CNS)*, Jun. 2019, pp. 187-195. <https://doi.org/10.1109/CNS.2019.8802732>
- [12] M. J. Page, J. E. McKenzie, P. M. Bossuyt et al., "The PRISMA 2020 statement: An updated guideline for reporting systematic reviews," *BMJ*, vol. 372, p. n71, Mar. 2021. <https://doi.org/10.1136/bmj.n71>
- [13] J. Lei, R. Jiang and Z. Wu, "Malicious ADS-B data generation based on improved GAN," in *IEEE BigDataSecurity/HPSC/IDS*, May 2023, pp. 72-77. <https://doi.org/10.1109/BigDataSecurity-HPSC-IDS58521.2023.00023>
- [14] W. Ahmed, A. Masood, J. Manzoor and S. Akleylek, "Automatic dependent surveillance-broadcast (ADS-B) anomalous messages and attack type detection: Deep learning-based architecture," *PeerJ Computer Science*, vol. 11, p. e2886, Jun. 2025. <https://doi.org/10.7717/peerj-cs.2886>
- [15] W. Ahmed, A. Masood and J. Manzoor, "Toward realistic and lightweight intrusion detection for ADS-B: A comprehensive threat model and multi-attack machine learning evaluation," *Array*, vol. 31, p. 101012, Sep. 2026. <https://doi.org/10.1016/j.array.2026.101012>
- [16] A. Brighente, M. Conti, S. Salaeva and F. Turrin, "One class to test them all: One-class classifier-based ADS-B location spoofing detection," in *Applied Cryptography and Network Security Workshops, LNCS*, vol. 14587, 2024, pp. 55-74. https://doi.org/10.1007/978-3-031-61489-7_4
- [17] N. Çevik and S. Akleylek, "Anomaly detection system for ADS-B data: Attack vectors and machine learning models," *Internet of Things*, vol. 29, p. 101416, Jan. 2025. <https://doi.org/10.1016/j.iot.2024.101416>
- [18] A. Fried and M. Last, "Facing airborne attacks on ADS-B data with autoencoders," *Computers & Security*, vol. 109, p. 102405, Oct. 2021. <https://doi.org/10.1016/j.cose.2021.102405>
- [19] A. Gurtov, J. D. Kenaudekar, S. Khan and F. L. Dos Santos De Souza, "Deep learning based anomaly detection for securing ADS-B in NextGen aviation," in *IEEE Aerospace Conference*, Mar. 2026. <https://doi.org/10.1109/AERO66936.2026.11519844>
- [20] E. Habler and A. Shabtai, "Analyzing sequences of airspace states to detect anomalous traffic conditions," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 58, no. 3, pp. 1843-1857, Jun. 2022. <https://doi.org/10.1109/TAES.2021.3124199>
- [21] R. Karam, M. Salomon and R. Couturier, "Supervised ADS-B anomaly detection using a

- false data generator,” in *2nd International Conference on Computer, Control and Robotics (ICCCR)*, Mar. 2022, pp. 218-223. <https://doi.org/10.1109/ICCCR54399.2022.9790149>
- [22] S. Khan, J. Thorn, A. Wahlgren and A. Gurtov, “Intrusion detection in automatic dependent surveillance-broadcast (ADS-B) with machine learning,” in *IEEE/AIAA 40th Digital Avionics Systems Conference (DASC)*, Oct. 2021. <https://doi.org/10.1109/DASC52595.2021.9594431>
- [23] R. Naaman, F. G. de Magalhaes, J.-Y. Ouattara and G. Nicolescu, “Quantum enhanced anomaly detection for ADS-B data using hybrid deep learning,” in *AIAA/IEEE 44th Digital Avionics Systems Conference (DASC)*, Sep. 2025. <https://doi.org/10.1109/DASC66011.2025.11257365>
- [24] M. Ngamboé, J.-S. Marrocco, J.-Y. Ouattara, J. M. Fernandez and G. Nicolescu, “New machine learning approaches for intrusion detection in ADS-B,” in *AIAA/IEEE 44th Digital Avionics Systems Conference (DASC)*, Sep. 2025. Author version: arXiv:2510.08333, <https://arxiv.org/pdf/2510.08333>.
- [25] H. Ould Slimane, S. Benouadah, K. Al Shamaileh, V. Devabhaktuni and N. Kaabouch, “ADS-B message injection attack on UAVs: Assessment of SVM-based detection techniques,” in *IEEE International Conference on Electro Information Technology (EIT)*, May 2022, pp. 405-410. <https://doi.org/10.1109/eIT53891.2022.9813819>
- [26] M. Pirolley, R. Couturier, M. Salomon and F. Ambert, “Machine learning techniques for detecting anomalies in ADS-B data in the context of low altitude aviation,” *Aerospace Science and Technology*, vol. 168, p. 111199, Jan. 2026. <https://doi.org/10.1016/j.ast.2025.111199>
- [27] M. A. Rahman, T. Bhuiyan and M. A. Ali, “Enhancing aviation safety: Machine learning for real-time ADS-B injection detection through advanced data analysis,” *Alexandria Engineering Journal*, vol. 126, pp. 262-276, Jul. 2025. <https://doi.org/10.1016/j.aej.2025.04.045>
- [28] V. Vajrobol, G. J. Saxena, S. Singh, A. Pundir, B. B. Gupta, A. Gaurav and K. T. Chui, “Enhancing aviation control security through ADS-B injection detection using ensemble meta-learning models with explainable AI,” *Alexandria Engineering Journal*, vol. 112, pp. 63-73, Jan. 2025. <https://doi.org/10.1016/j.aej.2024.10.042>
- [29] M. Yue, S. Shi, T. Gao and Z. Wu, “OCREN: A countermeasure for ADS-B attacks in the context of air traffic control,” *Expert Systems with Applications*, vol. 304, p. 130781, Apr. 2026. <https://doi.org/10.1016/j.eswa.2025.130781>
- [30] L. Zhong, X. Liu, H. Yang, Q. Zhong, J. Huang, T. Wu, W. Pan, K. Xu and M. Shang, “RDPA: Dual-domain deep learning for robust ADS-B spoofing detection via reversible normalization and spectral patch attention,” *Chinese Journal of Aeronautics*, p. 104083, Jan. 2026. <https://doi.org/10.1016/j.cja.2026.104083>
- [31] F. Shang, B. Wang, T. Li, J. Tian, K. Cao and R. Guo, “Adversarial examples on deep-learning-based ADS-B spoofing detection,” *IEEE Wireless Communications Letters*, vol. 9, no. 10, pp. 1734-1737, Oct. 2020. <https://doi.org/10.1109/LWC.2020.3002914>
- [32] T. Saito and M. Rehmsmeier, “The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets,” *PLOS ONE*, vol. 10, no. 3, p. e0118432, Mar. 2015. <https://doi.org/10.1371/journal.pone.0118432>